

VSOC Sprint Roadmap and Milestones

Purpose

This document outlines the structured development plan used to build the Virtual SOC environment. It documents deliverables, sequencing, and progress expectations across six distinct sprints. The roadmap supports academic tracking, project auditing, and repeatable rebuilds.

Sprint Structure Overview

The project is split into **six sprints**, each with specific goals leading to a functional SOC infrastructure: - Sprint 1 — Foundations and research - Sprint 2 — Network skeleton and initial deployment - Sprint 3 — NIDS implementation - Sprint 4 — SIEM & EDR integration - Sprint 5 — Full telemetry ingestion - Sprint 6 — Tuning, testing, and demonstration

Each sprint concludes with validated deliverables.

Sprint 1 — Architecture Foundations (Ends 9/21)

Goals

- Establish initial documentation structure
- Finalize diagrams (mermaid charts, high-level topology)
- Research all core technologies:
 - SIEM options
 - EDR frameworks
 - NIDS platforms
 - pfSense deployment models
- Prepare Git repository and issue board
- Prototype networks locally using VirtualBox while awaiting OpenStack access

Deliverables

- Initial architecture diagrams
 - Tooling research summary
 - Local lab sandbox
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Sprint 2 — Initial Cloud Deployment (Ends 10/05)

Goals

- Receive OpenStack access and begin real cloud deployment
- Build network skeleton:
 - ext → ex-router → transit-net → pfSense → LAN
- Deploy:
 - Kali attacker
 - pfSense firewall
 - Initial victim host
- Install Suricata package on pfSense
- Capture first traffic via NIDS
- Complete refined diagrams of end-of-semester SOC vision

Deliverables

- Fully routed pfSense instance
 - NIDS initial visibility test
 - Connectivity proven across all network segments
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Sprint 3 — NIDS Implementation and Tuning (Ends 10/19)

Goals

- Enable ET Open ruleset
- Verify Suricata visibility for all routed traffic
- Detect:
 - Nmap scans
 - HTTP traffic
- Test alert via `curl http://testmyids.com`
- Tune rules to reduce false positives
- Validate logging pathways within pfSense

Deliverables

- Working Suricata instance
 - ET rule updates functioning
 - First meaningful alerts captured
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Sprint 4 — SIEM & EDR Deployment (Ends 11/02)

Goals

- Install Wazuh Manager, Indexer, and Dashboard
- Confirm SIEM access over HTTPS
- Deploy Wazuh agents to multiple hosts
- Establish EDR telemetry collection
- Build rule correlation pipeline

Deliverables

- Fully functioning SIEM environment
 - Active endpoint agents
 - Initial correlation rules tested
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Sprint 5 — Log Forwarding & Pipeline Integration (Ends 11/16)

Goals

- Forward Suricata EVE JSON logs to Wazuh using syslog-ng
- Forward pfSense system logs into SIEM
- Validate:
 - NIDS → SIEM
 - Agents → SIEM
 - pfSense → SIEM
- Explore visualization tools such as ELK dashboards

Deliverables

- End-to-end telemetry ingestion
 - Suricata alerts visible in SIEM
 - Visualization prototypes
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Sprint 6 — Tuning, Automation, and Demonstration (Ends 12/07)

Goals

- Final alert tuning and noise reduction

- Prepare scripted demo workflow:
- nmap scans
- Metasploit workflows
- Malware simulation
- Construct an automated attack script (e.g., nmap + metasploit + curl)
- Build final presentation and documentation
- Validate SOC can detect all core attack techniques

Deliverables

- Clean detection pipeline with low false positives
- Full SOC demonstration
- Final academic and project documentation

Full Project Timeline Summary

Sprint	Focus Area	Key Outputs
1	Research, diagrams	Architecture plan, repo setup
2	Cloud deployment	pfSense, Kali, victim skeleton
3	Suricata NIDS	Ruleset + detection pipeline
4	Wazuh SIEM/EDR	Dashboard + agents
5	Syslog-ng + full ingestion	NIDS → SIEM pipeline
6	Tuning + demo	Final SOC showcase

Completion Status

All sprint objectives have been executed and validated successfully. The VSOC environment is now: - Fully operational - Instrumented with network + endpoint telemetry - Ready for demonstrations, coursework, and further research extensions